

SAGA, Japan

WMO: 478130

Lat: 33.2664N

Lon: 130.3053E

Elev: 108

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.3	32.1	16.8	12.9	41.7	19.7	14.9	41.5	23.2	45.4	20.9	44.7	4.9	70	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.7	94.5	77.8	92.5	77.5	90.5	77.0	80.5	88.9	79.6	87.5	78.7	86.3	8.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.5	148.7	83.9	77.6	143.9	83.4	76.6	139.4	83.0	44.2	89.1	43.2	87.6	42.3	86.5	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.1	18.7	16.4	DB	26.3	97.9	1.9	2.0	24.9	99.3	23.8	100.5	22.8	101.6	21.4	103.0	
			WB	24.6	81.3	1.7	1.5	23.4	82.4	22.4	83.3	21.4	84.1	20.2	85.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	42.5	45.0	50.8	59.7	68.7	75.0	82.1	83.8	77.1	67.3	56.3	46.4
	DBStd	15.02	4.63	5.62	5.57	5.47	4.14	3.71	4.01	3.45	4.57	5.08	6.02	5.51
	HDD50	611	238	161	57	3	0	0	0	0	0	0	12	140
	HDD65	2761	699	561	442	171	14	0	0	0	0	33	265	576
	CDD50	5355	4	20	81	295	580	749	995	1048	814	538	202	29
	CDD65	2029	0	0	0	13	128	299	530	583	365	106	5	0
	CDH74	18666	0	0	0	71	783	1937	5742	6782	2863	478	10	0
	CDH80	6893	0	0	0	4	144	479	2325	2980	898	63	0	0
Wind	WSAvg	6.9	6.4	6.8	7.4	7.0	6.6	7.0	7.4	7.2	7.1	7.0	6.1	6.2
Precipitation	PrecAvg	71.0	2.6	3.1	4.4	5.6	6.5	11.9	12.1	8.4	6.7	3.4	3.8	2.6
	PrecMax	99.5	7.4	5.6	10.3	10.9	13.4	20.1	22.9	22.6	22.1	7.9	7.4	5.4
	PrecMin	37.4	1.1	1.1	2.2	1.9	1.7	1.4	0.7	1.6	2.5	0.4	0.6	0.4
	PrecStd	13.5	1.3	1.3	1.7	2.4	3.1	5.5	6.7	5.0	4.1	2.2	1.9	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	66.0	71.5	79.6	87.3	89.8	96.9	97.2	93.0	84.8	75.5	64.4
		MCWB	52.5	56.6	58.6	62.0	64.8	70.7	79.0	78.5	76.2	71.3	63.5	56.3
	2%	DB	55.9	61.6	67.6	75.8	83.6	87.3	94.1	95.1	90.3	81.6	71.8	61.1
		MCWB	49.4	53.8	56.1	60.8	64.8	70.6	77.8	78.1	75.5	68.5	61.7	53.3
	5%	DB	53.4	57.9	64.2	73.0	81.1	85.0	92.1	93.4	87.9	79.0	68.9	58.5
		MCWB	47.2	50.6	54.7	59.4	64.1	70.8	77.6	77.9	74.5	67.2	60.1	52.0
	10%	DB	51.1	54.8	61.0	70.2	78.6	82.8	89.9	91.4	85.2	76.5	66.4	56.0
		MCWB	45.0	48.0	52.8	59.1	63.6	70.8	77.5	77.7	73.4	65.7	58.9	50.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.7	59.6	62.1	69.0	72.9	78.0	81.7	81.9	79.6	75.6	67.7	59.6
		MCDB	57.6	62.6	65.9	72.8	77.4	82.7	90.8	90.2	86.6	80.7	71.8	60.8
	2%	WB	51.5	55.8	59.2	65.8	70.9	76.6	80.4	80.8	78.0	72.8	64.6	55.7
		MCDB	53.9	59.6	64.1	70.6	74.9	81.7	89.0	89.1	85.1	78.5	68.0	59.0
	5%	WB	48.2	52.3	56.7	63.7	68.9	75.5	79.5	79.9	76.8	70.2	62.4	52.8
		MCDB	52.1	56.3	62.4	69.1	74.4	80.5	87.3	88.1	83.9	76.4	67.1	56.9
	10%	WB	45.7	49.2	54.3	61.4	67.0	74.2	78.6	79.1	75.6	67.6	60.1	50.5
		MCDB	50.4	53.8	59.9	67.5	74.2	79.0	85.9	87.1	82.4	73.4	65.5	55.4
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.8	16.0	17.3	17.5	13.3	12.4	13.7	14.0	15.8	15.3	14.1
		MCDBR	16.7	18.8	20.4	21.9	22.4	18.0	16.3	16.6	16.9	18.2	17.4	16.8
		MCWBR	11.3	12.2	12.4	10.9	9.5	6.8	5.5	5.4	6.5	8.1	9.5	10.9
	5% WB	MCDBR	14.0	15.0	15.9	14.9	13.6	10.2	13.6	14.0	13.4	13.2	13.3	13.8
		MCWBR	11.5	12.4	12.2	9.9	8.5	5.2	5.2	5.3	5.8	7.4	9.2	10.8
Clear-Sky Solar Irradiance	taub	0.429	0.470	0.529	0.548	0.557	0.571	0.561	0.533	0.504	0.457	0.428	0.421	
	taud	2.061	1.989	1.845	1.853	1.892	1.942	2.013	2.088	2.124	2.158	2.152	2.096	
	Ebn at Noon	238	242	238	241	240	236	237	242	243	244	238	232	
	Edh at Noon	42	50	62	65	63	60	56	51	47	42	38	39	
All-Sky Solar Radiation	RadAvg	697	924	1267	1524	1645	1388	1555	1632	1324	1119	827	647	
	RadStd	68	89	84	131	188	187	224	167	137	115	80	46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+178	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page